

Alejandro Gomez De Mendieta

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EDUCATION

University of Texas at Dallas

Bachelor of Science in Computer Science, Minor in Statistics

Expected Dec 2027

Relevant Coursework: Systems Programming in UNIX, Data Structures and Algorithms, Advanced Algorithm Design, Probability and Statistics, Advanced Calculus

Austin Community College

Associate of Science in Computer Science; Associate of Science in Mathematics

2025

SKILLS

Languages: C++, C, Python, Java, R

Systems and Tools: Linux/UNIX, POSIX sockets, TCP/IP, CMake, Git, Catch2, CTest, GitHub Actions, Docker, Redis, Elasticsearch

Certifications: CompTIA A+, CompTIA Network+

EXPERIENCE

Software Engineer (Part-time), Ohel Technologies

2022 - 2025

Python, REST APIs, Elasticsearch

- Automated internal data and IT workflows for a five-person team and documented them in five SOPs, reducing recurring manual work by approximately five hours per week.
- Engineered biweekly ingestion pipelines integrating five external APIs with Elasticsearch, loading 100,000+ records in three months to support an AI application.
- Designed quota-aware ingestion logic around third-party rate limits and monthly usage caps to maintain reliable data collection.

Computer Science II Course Grader, University of Texas at Dallas

Jan 2026 - Present

Java, Code Review

- Grade approximately 30 Java submissions per week for 71 students across two sections, maintaining a typical three-day turnaround against a seven-day policy.
- Provide rubric-aligned feedback on correctness, edge cases, code quality, and specification compliance to help students diagnose logic errors and improve implementations.
- Maintain shared notes on recurring mistakes and ambiguous edge cases to improve grading consistency across two graders.

PROJECTS

Modern C++ HTTP Server

2026

C++23, Linux/POSIX sockets, CMake, Catch2, CTest, GitHub Actions

- Built a from-scratch HTTP/1.1 server in C++23 using Linux/POSIX sockets, move-only RAII file descriptor ownership, layered TCP/HTTP abstractions, and method/path-based request routing.
- Implemented an incremental HTTP parser supporting fragmented socket reads, request-line/header/body validation, malformed-request handling, and response serialization; validated correctness with 111 Catch2 unit and process-level integration tests over real TCP sockets.
- Built a custom nanosecond-resolution benchmarking framework with per-iteration sampling and statistical analysis including mean, median, standard deviation, p95/p99 latency, throughput, and measurement-overhead correction across workloads of up to 100,000 iterations.
- Profiled parser workloads with Linux *perf*, traced ~85% of sampled self CPU to repeated buffer movement in *memmove*, and redesigned parsing around a cursor-based buffer, reducing byte-by-byte request parsing from ~414 ns to ~162 ns.

Homefull, HackUTD

2025

Python, NVIDIA Nemotron, DigitalOcean

- Deployed NVIDIA Nemotron on a DigitalOcean GPU droplet and built a Python backend that transformed crawled webpage content into structured JSON for local resource and event discovery.
- Implemented URL fetching, HTML parsing, event extraction, geocoding, and deduplication across the top five search results and pages crawled within each domain.
- Integrated an end-to-end discovery pipeline that returned event time and location information to the mobile application.

RESEARCH

Student Researcher, The Data Mine, Purdue University

Jan 2026 - Present

Modularized Space Vehicle Self-Recovery From Cyber Threats - U.S. Space Force

- Review two technical papers per week on cyber-resilient spacecraft and synthesize findings to support the design of resilient modularized space vehicles.
- Contribute to a Python and pandas detection prototype that flags unusual activity across spacecraft modules; collaborate through weekly sprints to deliver a prototype, poster, presentation, and technical report.

LEADERSHIP AND ACTIVITIES

ColorStack at UTD

Tech Coordinator

Jul 2026 - Present

F1 Club at UTD

Onboard Car Software Team

2025 - 2026

CodePath

TIP102 Technical Interview Preparation

Spring 2026

Baking Club at UTD

Co-founder and Vice President

2025 - Present